The sensitivity of the child to sex steroids: possible impact of exogenous estrogens.

The current trends of increasing incidences of testis, breast and prostate cancers are poorly understood, although it is assumed that sex hormones play a role. Disrupted sex hormone action is also believed to be involved in the increased occurrence of genital abnormalities among newborn boys and precocious puberty in girls. In this article, recent literature on sex steroid levels and their physiological roles during childhood is reviewed. It is concluded that (i) circulating levels of estradiol in prepubertal children are lower than originally claimed; (ii) children are extremely sensitive to estradiol and may respond with increased growth and/or breast development even at serum levels below the current detection limits; (iii) no threshold has been established, below which no hormonal effects can be seen in children exposed to exogenous steroids or endocrine disruptors; (iv) changes in hormone levels during fetal and prepubertal development may have severe effects in adult life and (v) the daily production rates of sex steroids in children estimated by the Food and Drug Administration in 1999 and still used in risk assessments are highly overestimated and should be revised. Because no lower threshold for estrogenic action has been established, caution should be taken to avoid unnecessary exposure of fetuses and children to exogenous sex steroids and endocrine disruptors, even at very low levels.